Machine Learning for (CA1)

DUBLIN PUBLIC TRANSPORT ANALYTICS & FORECASTING

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INTRODUCTION

- Analyzed Dublin public transport passenger data (dem-public-transport-journeys.csv).
- Goal: Group travel patterns and forecast future passenger trips.
- Real-world context: Examined issues like DART train breakdowns and passenger reliance on the H3 bus route.



CLUSTERING ANALYSIS (K-MEANS & DBSCAN

- Standardized numeric data using StandardScaler.
- K-Means (K=3): Successfully grouped days into Low demand, Normal, and Peak busy days.
- DBSCAN: Useful for detecting transportation delay anomalies and unexpected disruptions.

TiME SERiES FORECASTiNG (ARIMA)

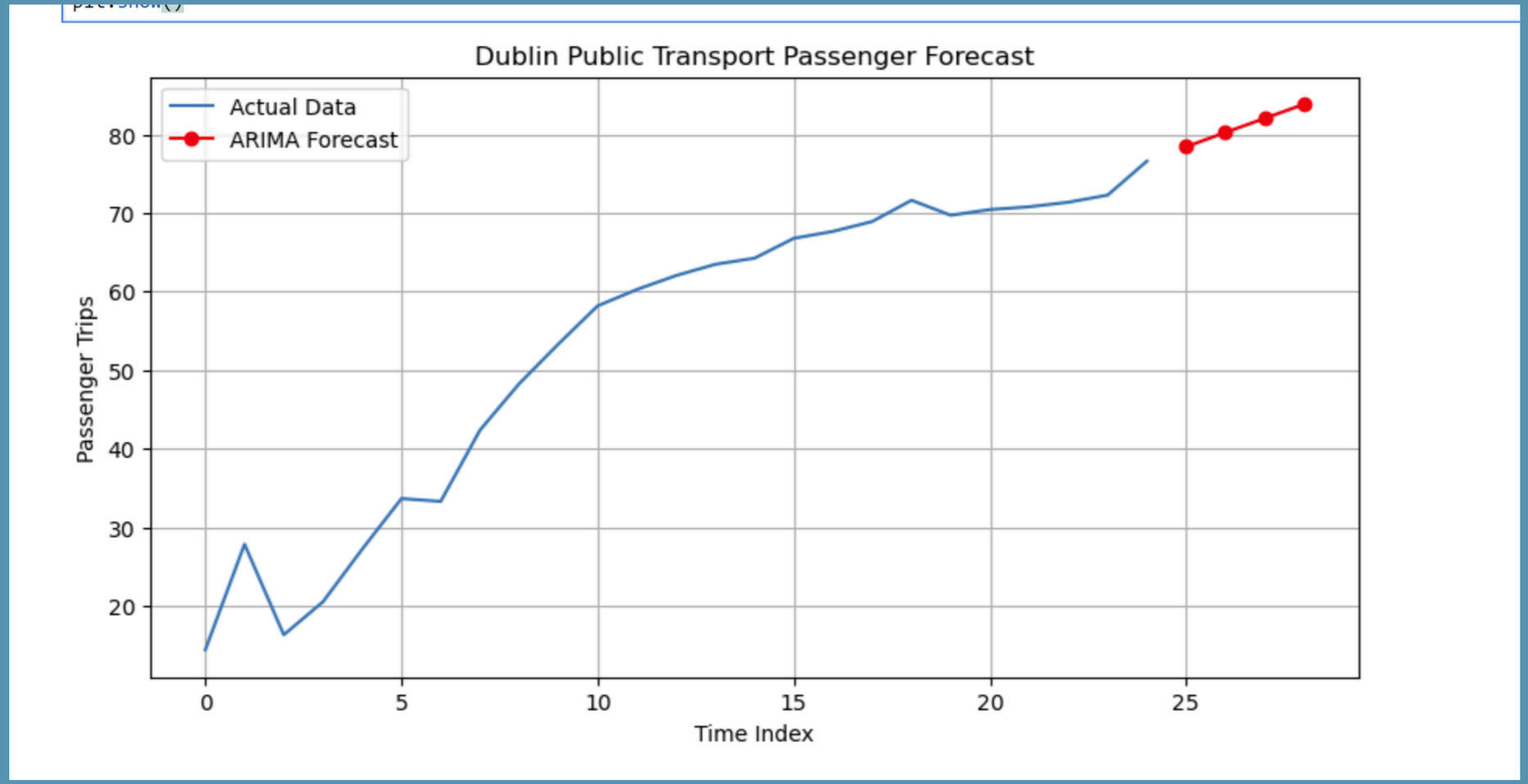


Implemented ARIMA(1, 1, 1) model to predict short-t
passenger numbers.
Used differencing (d=1) to achieve stationarity.
Generated a 4-step future passenger forecast.



CONCLUSION

- K-Means effectively segments daily passenger demand.
- ARIMA helps transport managers forecast trips and schedule extra H3 buses during train disruptions.


Dublin Public Transport Passenger Forecast
Actual Data
ARIMA Forecast
Passenger Trips
Time Index
20
30
40
50
60
70
80
0
5
10
15
20
25